technische universität
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Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Quantitative Process Analysis

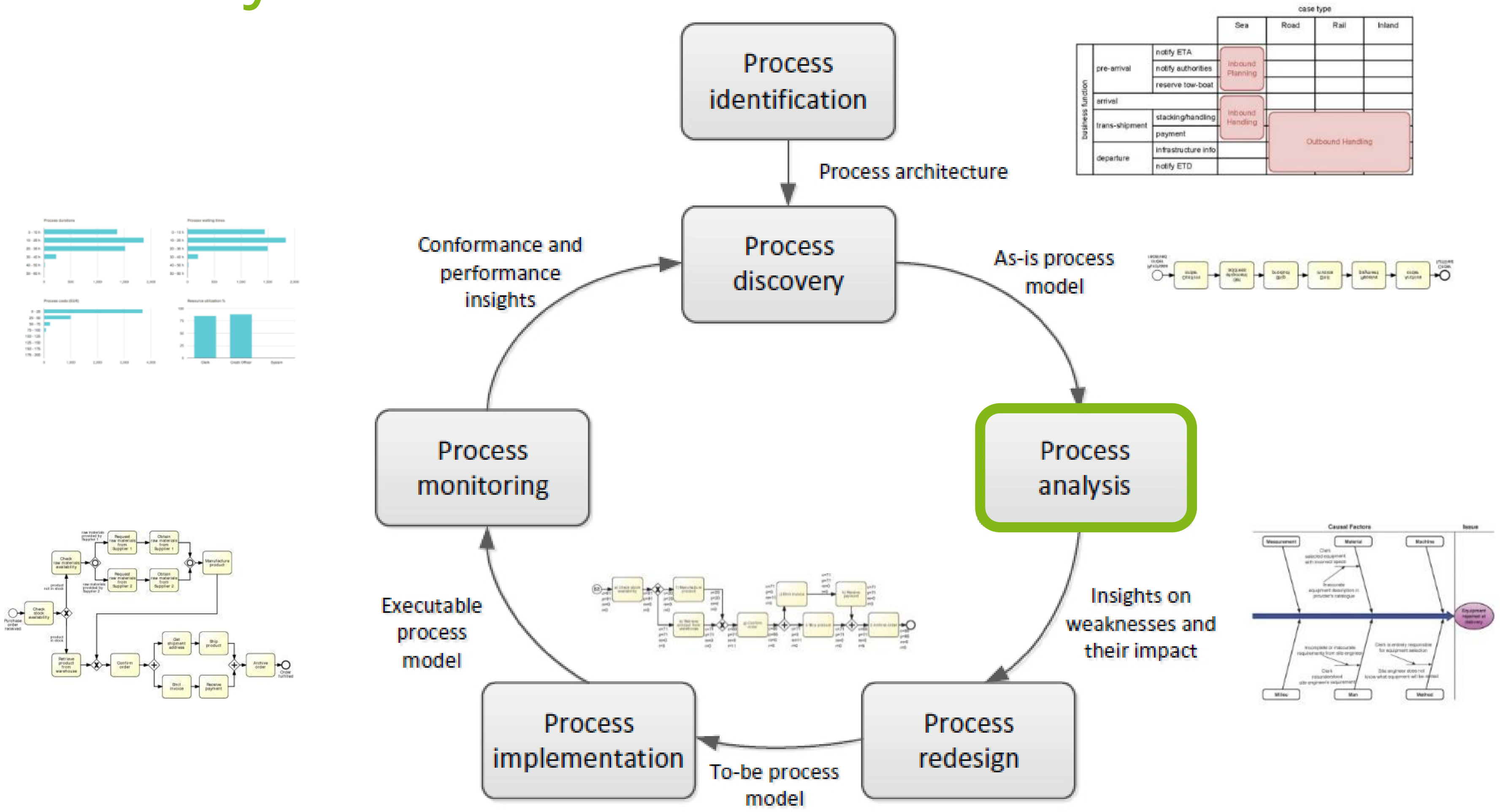
Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

- Understand process performance measures
- Understand and be able to use flow analysis
- Understand and be able to use queueing analysis
- Understand process simulation

The BPM Lifecycle



Dumas et al. (2018)

Process Analysis

“Quality is free, but only to those who are willing to pay heavily for it.”

Tom DeMarco (1940–)

- Analyzing Processes is an **art** and a **science**
- **Qualitative analyses** use a range of **principles and techniques**
- **Quantitative analyses** use **measures** to judge a process
- There is no single way of producing good results

Dumas et al. (2018)

A Selection of Process Analysis Techniques

- **Qualitative** analysis
  - Value-added Analysis
  - Waste Analysis
  - Stakeholder Analysis
  - Root cause Analysis

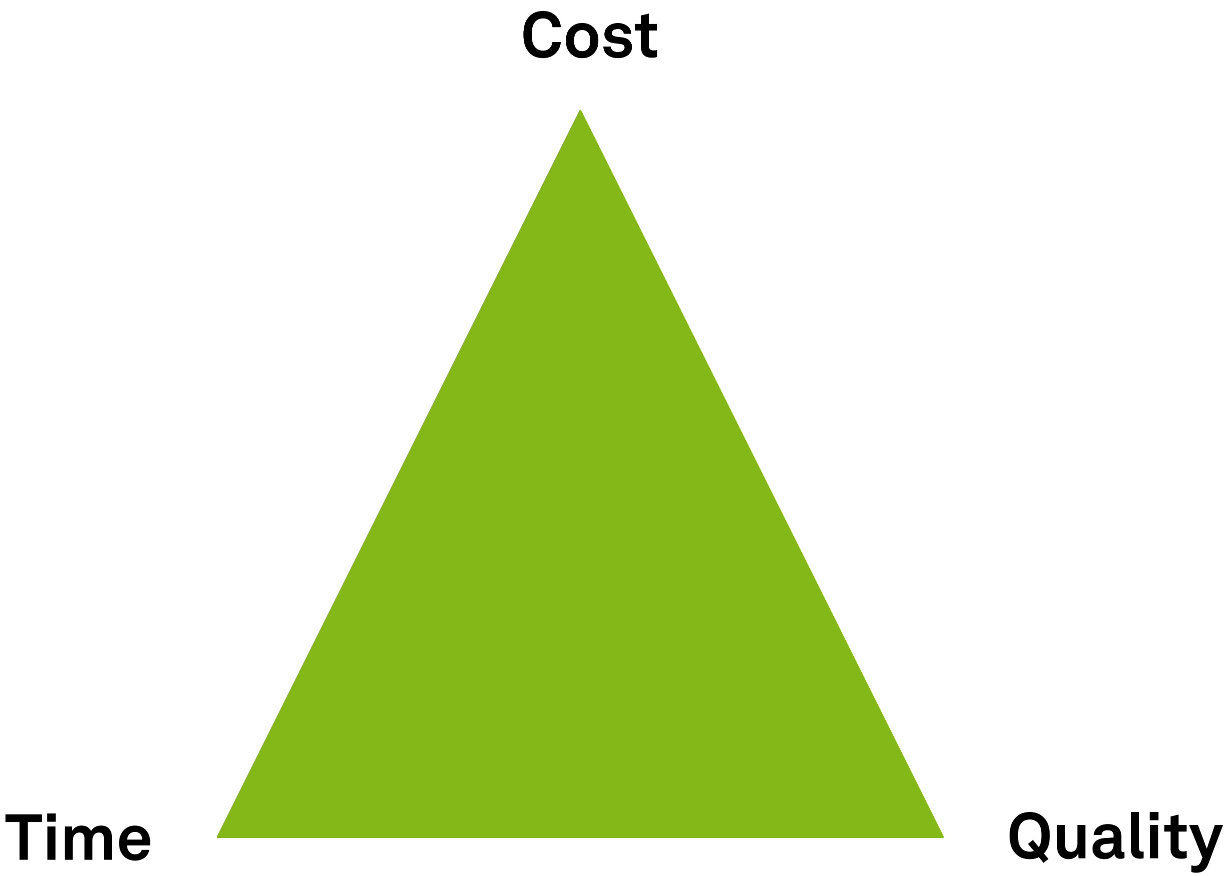
- **Quantitative** Analysis
  - Flow Analysis
  - Queueing Analysis
  - Process Simulation

Dumas et al. (2018)

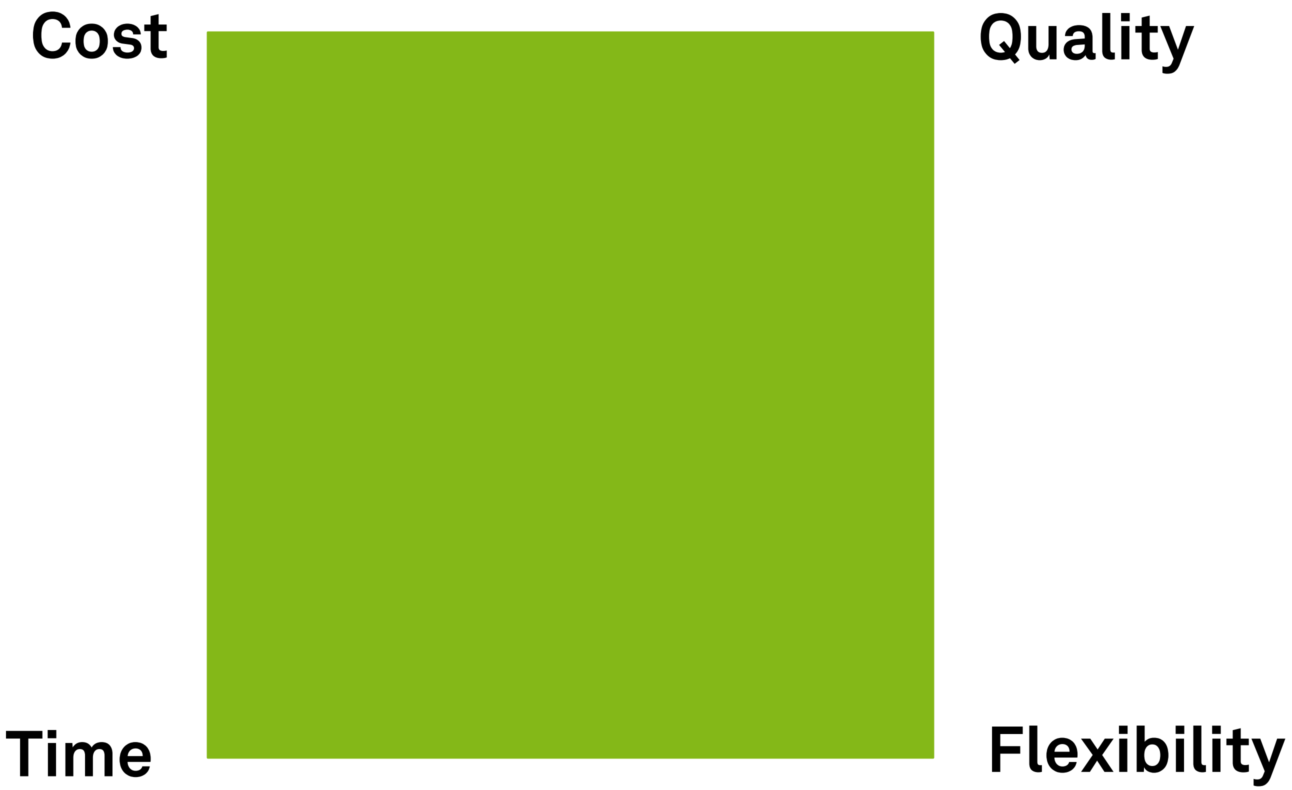
Process Performance Measures




Performance Dimensions

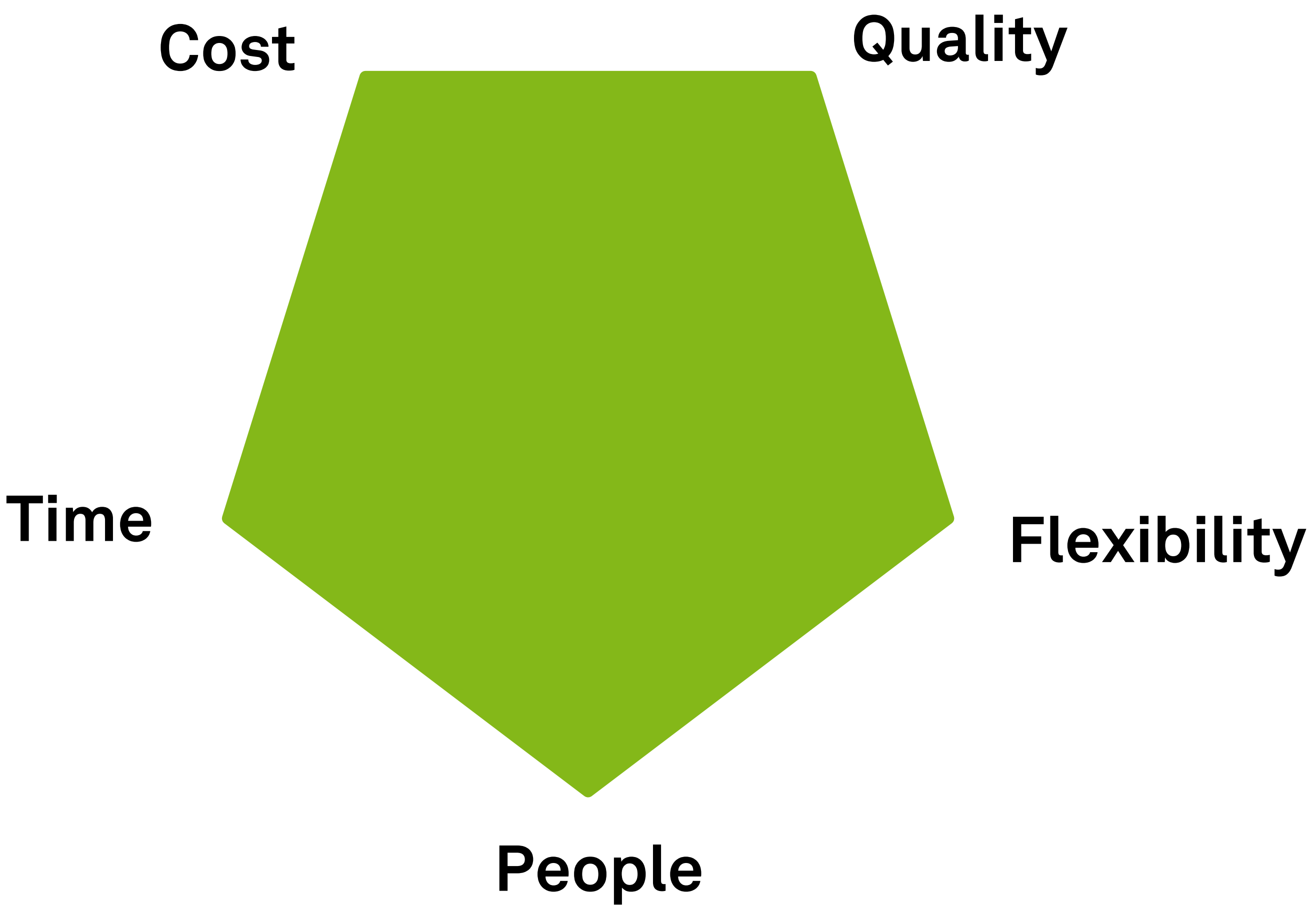


The Devil's Quadrangle: Process Performance Dimensions



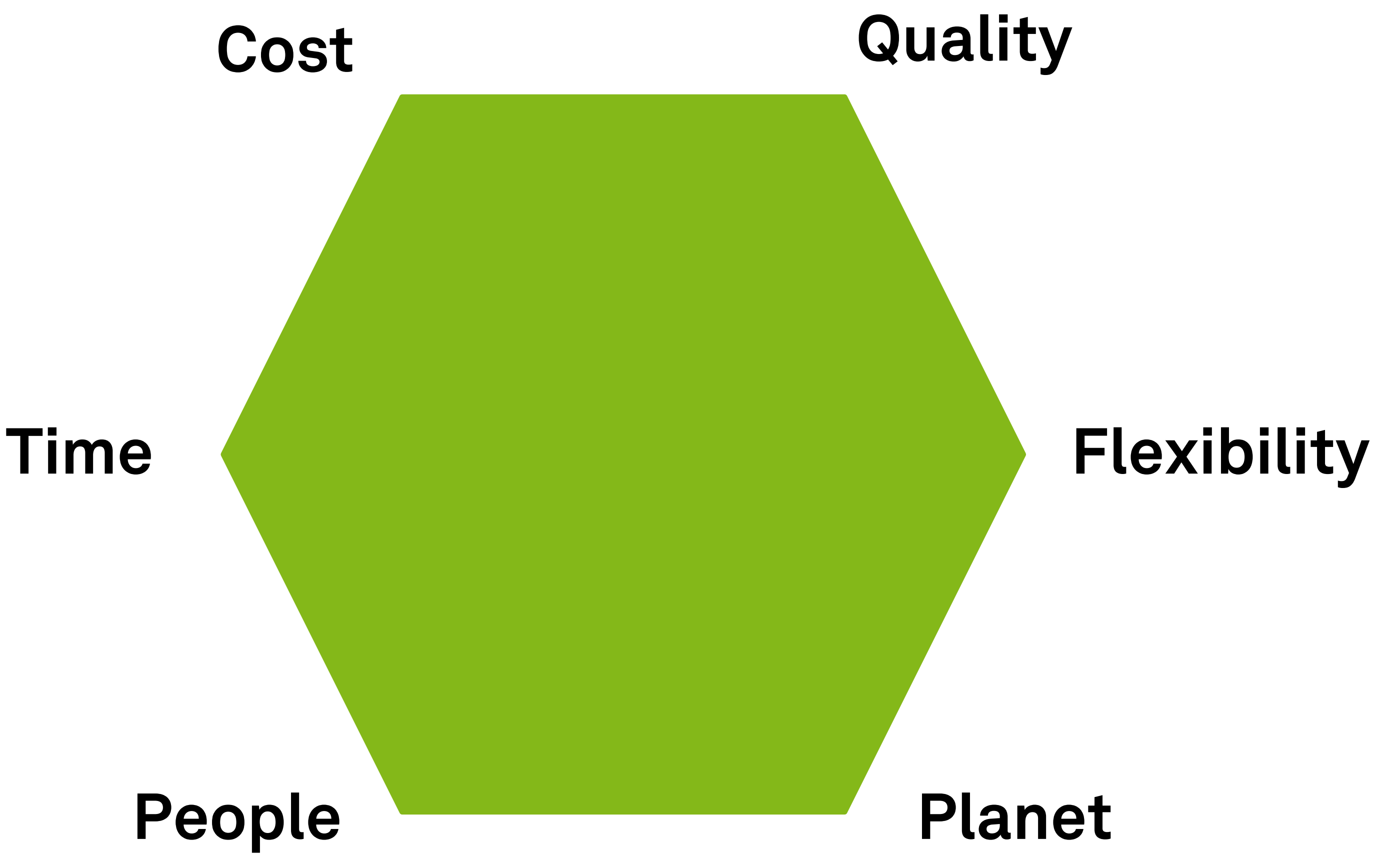
Dumas et al. (2018)

Goal Pentagon: More Process Performance Dimensions



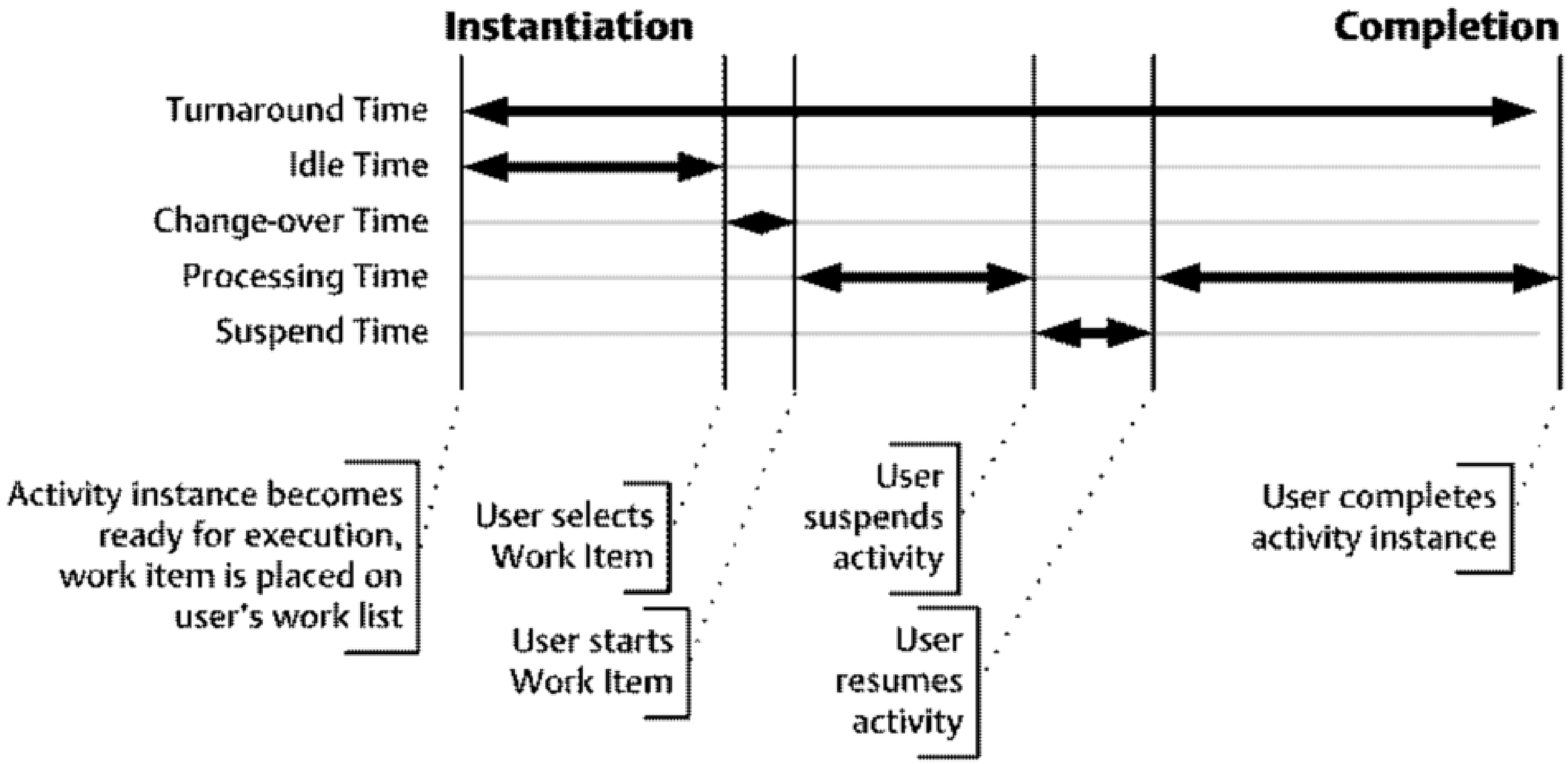
Francois et al. (2022)

Triple Bottom Line: More Performance Dimensions



Dumas et al. (2018), Elkington (2018), François et al. (2022)

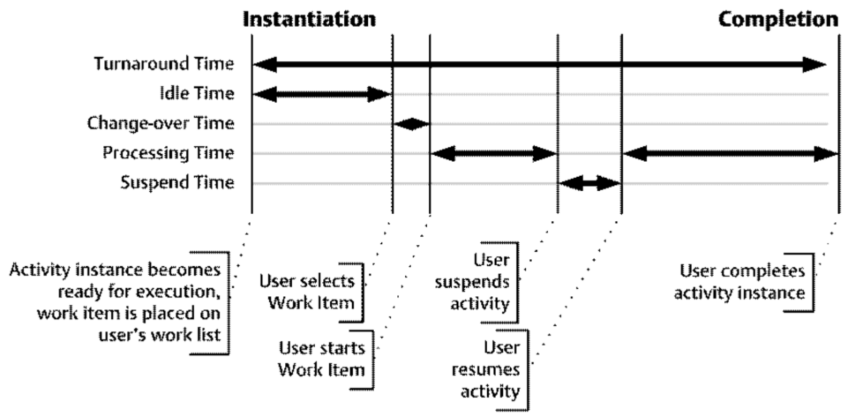
Time Data of a Process



zur Mühlen (2004)

Process Performance Measure: Time

- **Cycle time/ throughput time**
  – Time it takes to complete a process
- **Processing time**
  – Time a resource works on the process
- **Waiting time**
  – Time a resource does not work on the process
- **Time spent in non-value-adding tasks**
  – Time a resource does not work on value-adding activities in the process



– Typical scopes
  - average cycle time
  - maximal cycle time
  - meet service level agreement (SLA) time
  - find variation

Dumas et al. (2018)

Process Performance Measure: Cost

- Could also be turnover, revenue, or yield

- **Fixed**
  - Overhead cost that are hard to change
    - infrastructure and maintenance cost (utilization)

  → activity-based costing

- **Variable**
  - Operational cost related to productivity/ input & output
    - per execution
    - in particular labor cost and software services

Dumas et al. (2018)

Process Performance Measure: Quality

- **Internal quality**
  - As perceived by internal participants
    - level of control
    - level of variation/variants
    - context of work
    - errors
- **External quality**
  - As perceived by the client/ customer
  - Information the customer receives
    - Amount, relevance, quality, timeliness (could be time dimension as well)
  - e.g., churn rate or net promoter score

Dumas et al. (2018)

Process Performance Measure: Flexibility

- Ability to **react to changes**
- Flexibility of
  - **resources** (ability to execute many tasks/ new tasks)
  - **process** (ability to handle various cases and changing workloads)
  - **management** (ability to change rules/ allocation)
  - **organization** (ability to change the structure and responsiveness to demands of market or business partners

- **Run time** vs. **build time** flexibility

Dumas et al. (2018)

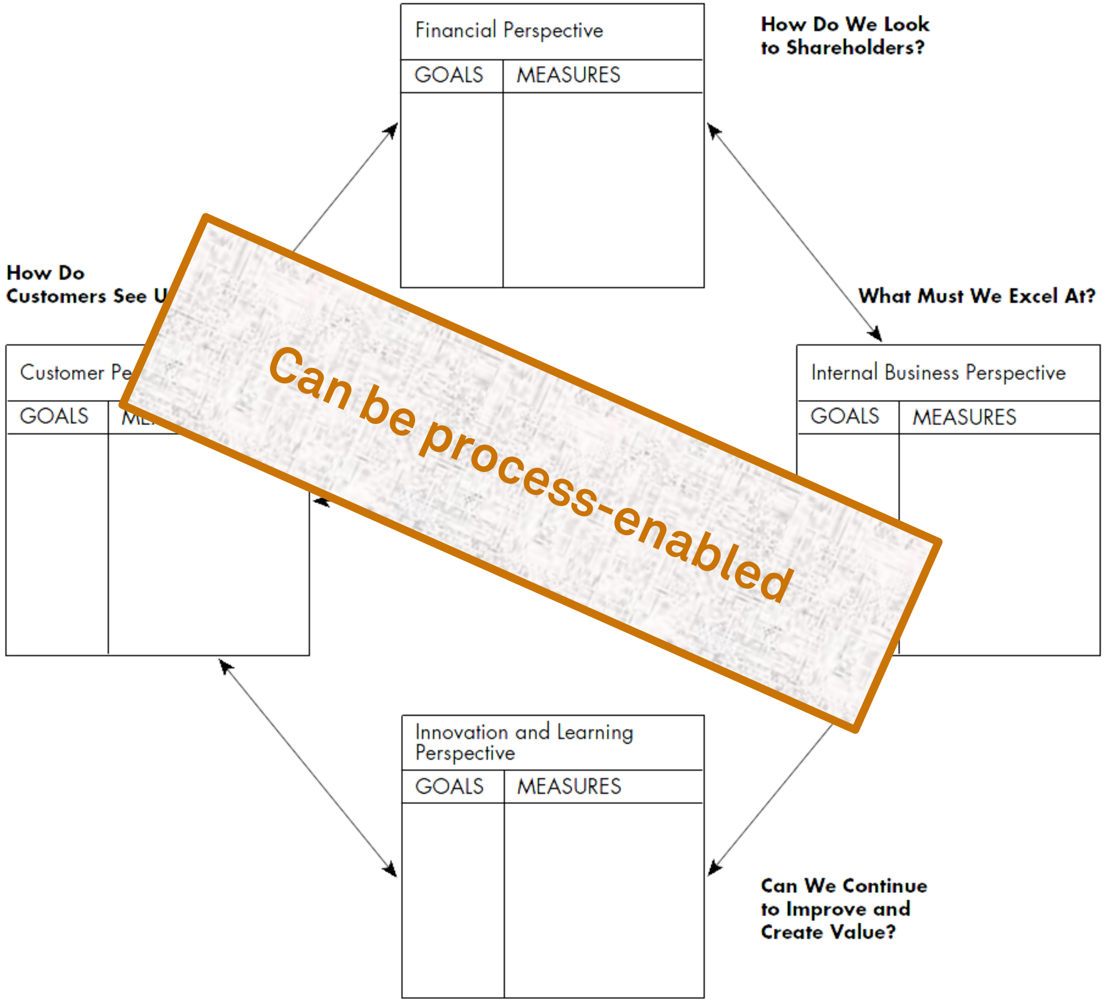
Definition of Process Performance Measures

1. Formulate **performance objectives** of the process at a high level, in the form of a **desirable state** that the process should ideally reach
   – "customers should be served in less than 30 minutes"
2. For each performance objective, identify the relevant **performance dimension(s)** and **aggregation function(s)**, and from there, define one or more **performance measures** for the objective in question
   – "percentage of customers served in less than 30 minutes"
3. Refine this performance measure using a **performance target** and attaching a **timeframe**
   – "99% of customers served in less than 30 minutes until 31/12/2023"

Dumas et al. (2018)

Balanced Scorecard

- Focuses on the **strategic agenda** of the organization
- Focused set of **measures** to monitor performance against **objectives**
- Mix of **financial** and **non-financial** data items
  - Originally: financial, customer, internal process, learning & growth
- Portfolio of initiatives to impact performance



Kaplan, Norton (1992)

Measure Derivation via Reference Models

- Reference model is used as a template to design the process performance measures

- Examples:
  - Process Classification Framework (PCF)
  - Information Technology Infrastructure Library (ITIL)
  - Supply Chain Operations Reference Model (SCOR)

Dumas et al. (2018)

Flow Analysis

Flow Analysis

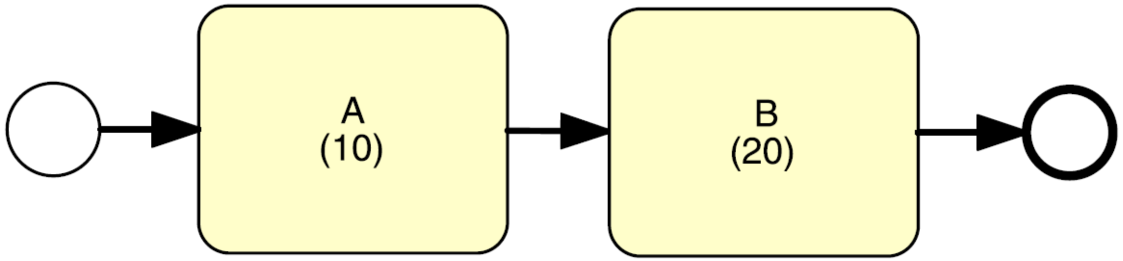
- Family of techniques to estimate **overall performance** based on some knowledge about the performance of its **tasks**

- **Cycle time analysis** calculates the average cycle time for a process or a process fragment

- Recap
  - Cycle time: Difference between a process start and end time

Dumas et al. (2018)

Performance of a Sequence



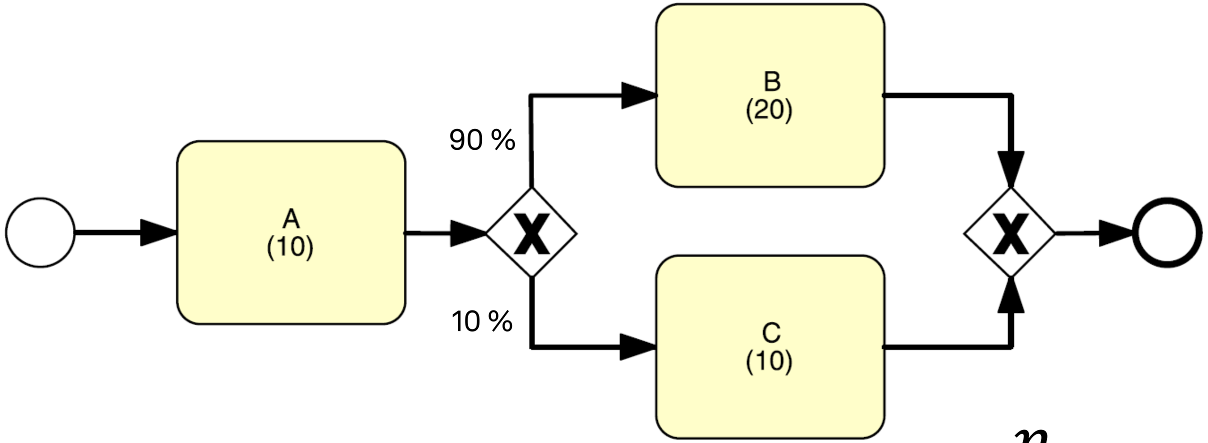
A + B = 10 + 20 = 30

$$CT = \sum_{i=1}^{n} T_i$$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a XOR Block



$A + p_1 * B + p_2 * C =$
$10 + 0.9 * 20 + 0.1 * 10 = 29$

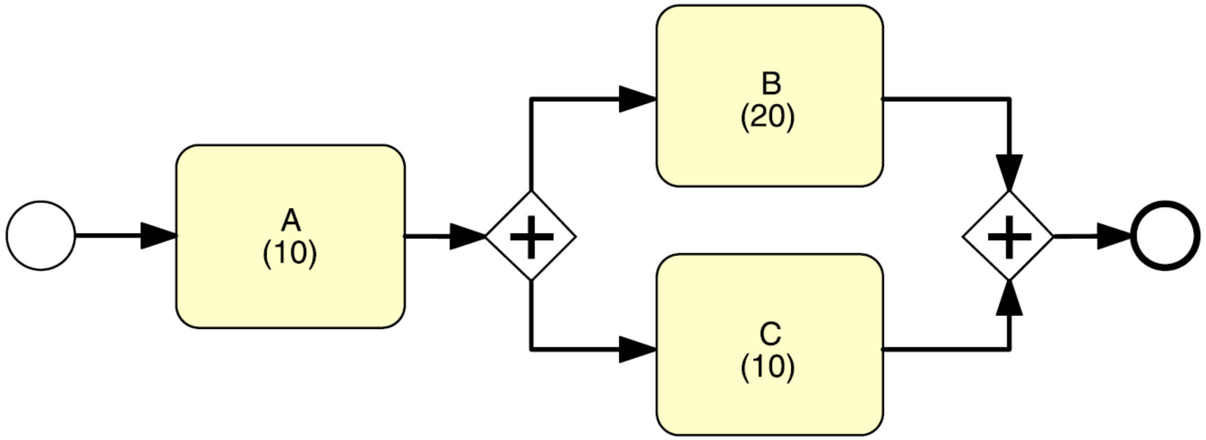
$$CT = \sum_{i=1}^{n} p_i \times T_i$$

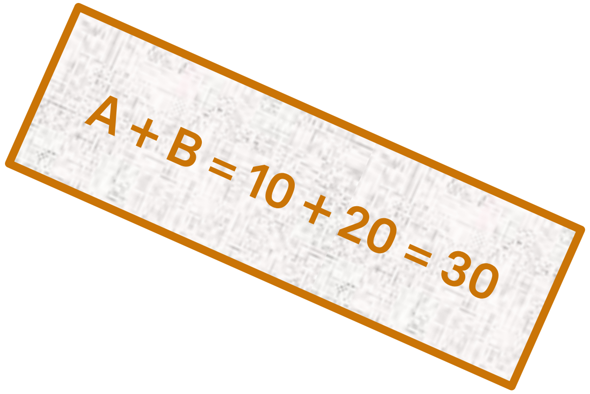
$p_i = branching\ probability\ of\ a\ process\ fragment$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a AND Block



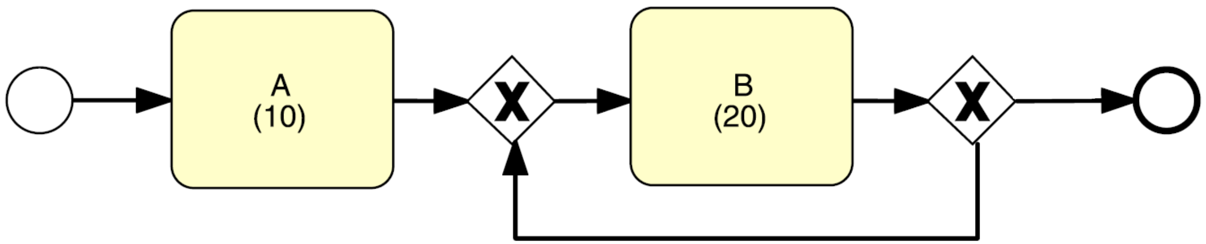


$$CT = Max(T_1, T_2, \ldots, T_n)$$

$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a Loop Block (Rework)



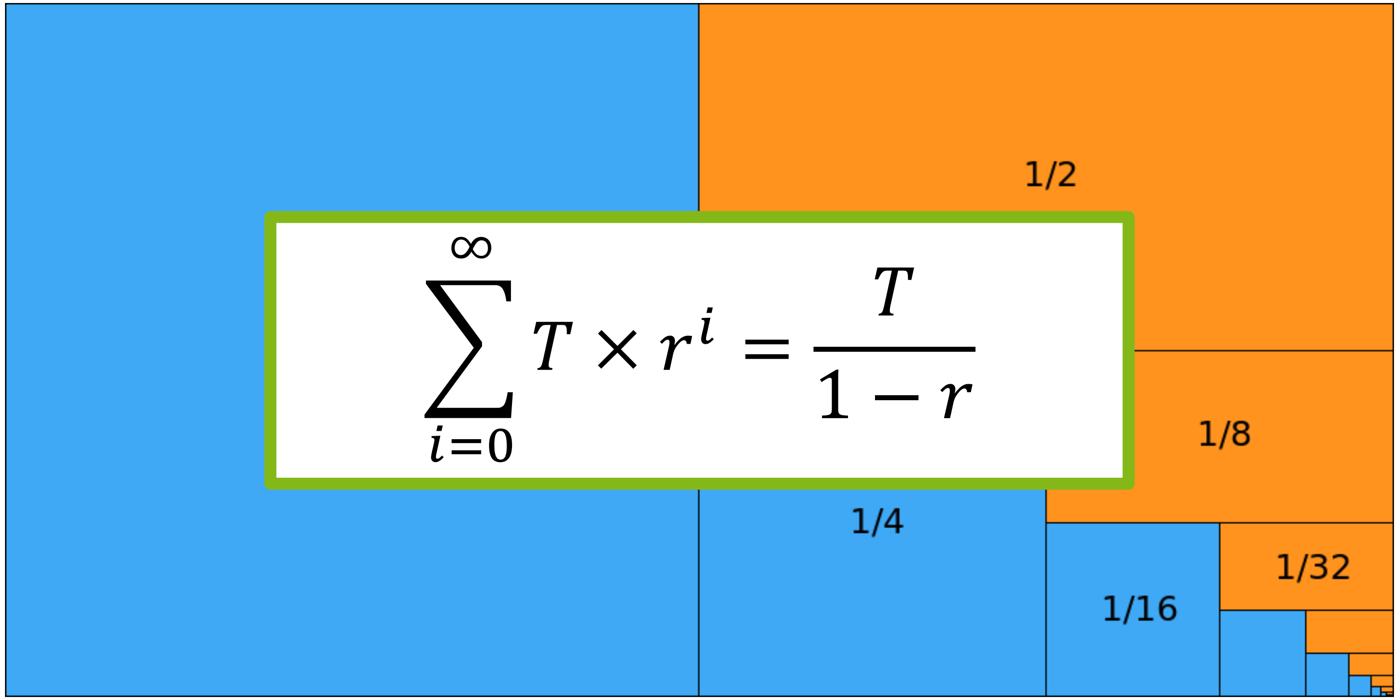
A + B + B*r + B*r² + … + B*r^N =
10 + 20 + 4 + 0,8 + … = ???

$$CT = \sum_{i=0}^{\infty} T \times r^{i}$$

$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Geometric Series (50 % Example)

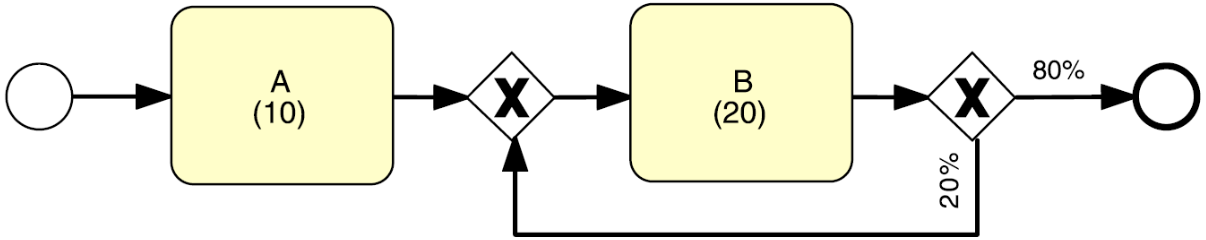


$$\sum_{i=0}^{\infty} T \times r^i = \frac{T}{1-r}$$

$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Dumas et al. (2018)

Performance of a Loop Block (Rework)



A + B/(1-r) = 10 + 20/(1-0,2) = 35

$$CT = \frac{T}{1-r}$$

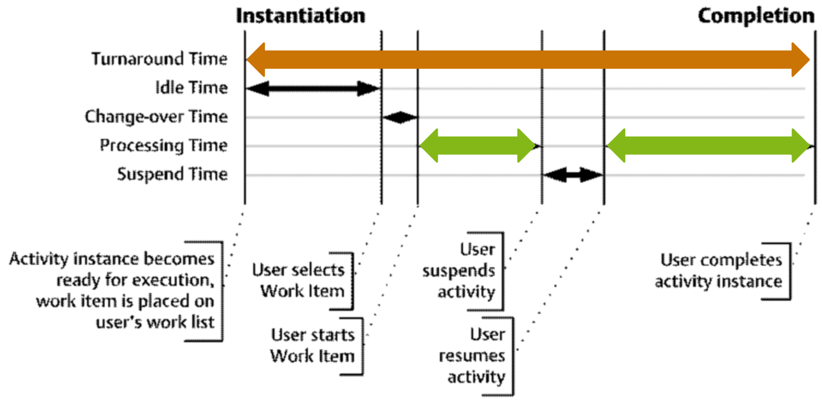
$r = rework\ probability$
$T_i = cycle\ time\ of\ a\ process\ fragment$
$CT = cycle\ time\ of\ a\ process$

Assumes uniform probabilities and durations for every rework loop

Dumas et al. (2018)

Cycle Time Efficiency

- Ratio of **theoretical processing time relative to overall processing time**
- Theoretical cycle time does not have waiting/ idle time



$$CTE = \frac{TCT}{CT}$$

$TCT = theoretical\ cycle\ time$

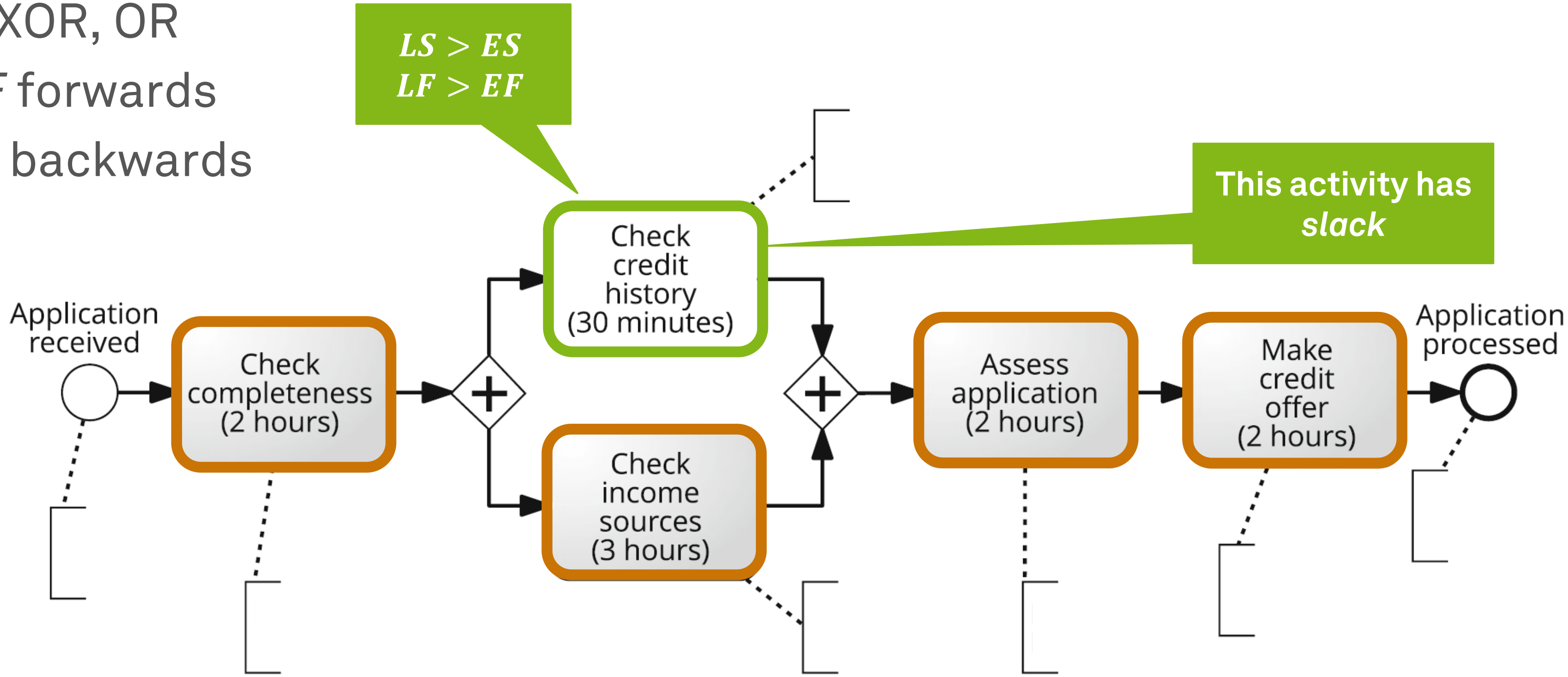
$CT = cycle\ time\ of\ a\ process$

$CTE = cycle\ time\ efficiency$

Dumas et al. (2018)

Critical Path Method

- Identify tasks that contribute to the theoretical cycle time
  - Remove or simplify XOR, OR
  - Calculate *ES* and *EF* forwards
  - Calculate *LS* and *LF* backwards



$ES = early\ start$
$EF = early\ finish$
$LS = late\ start$
$LF = late\ finish$

Dumas et al. (2018)

Little's Law for What-If Analysis

- Average **cycle time of a process**
- Average number of **new process instances per time unit** (arrival rate)
- Average number of **process instances active at a given time** (work-in-progress, started but not completed)
  - → If cycle time or arrival rate increases, WIP increases, and the process slows down
  - → If the arrival rate increases, the cycle time must decrease or the WIP increases as well

$$WIP = \lambda \times CT$$

Holds for any stable process

$CT = cycle\ time\ of\ a\ process$
$\lambda = arrival\ rate$
$WIP = work{-}in{-}process$

Arrival rate can be substituted by throughput (average completed instances per time unit)

Dumas et al. (2018)

Capacity and Bottlenecks

- **Theoretical capacity** is the maximum number of instances that can be completed per time unit given a set of resources

$$\mu_p = \frac{uc}{ul}$$

$p = resource\ pool$
$uc = unit\ capacity$
$ul = unit\ load$

- **Resource utilization** (occupation rate) of a pool is the share of working vs. idle workers

$$\rho_p = \frac{\lambda}{\mu_p}$$

$\lambda = arrival\ rate$

- The pool that reaches 100% resource utilization first is called a **bottleneck** and determines the overall theoretical capacity of the process

Dumas et al. (2018)

Summary Flow Analysis

- Flow Analysis is not limited to **time** analysis
- Can be used to estimate **error rates** or **capacity requirements**
- Can be used to estimate **cost**
  - Sequence, XOR, loop are calculated similar
  - AND is calculated like a sequence
- **Limitations**
  - Does not work on any process model (blocks required)
  - OR gateways are more complicated
  - Average cycle times must be known (interviews/ observations/ logs)
  - Load behavior (resource contention)

    → **When resource utilization > 90%, waiting time increases steeply**

Dumas et al. (2018)

Queueing Analysis

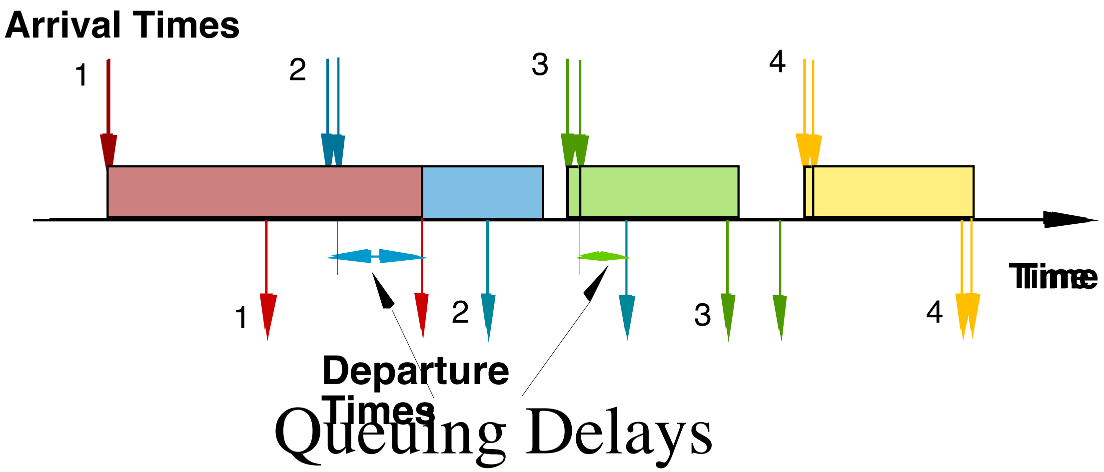
Reasons for Contention

- Not all traffic is deterministic
- Arrival rate is just an average
- Variable but **spaced apart** traffic
- **Burstiness** (variability in service times and/or inter-arrival times)
- **Job size** variation
- **High utilization** interference

Dumas et al. (2018)

Motivation Queueing Analysis

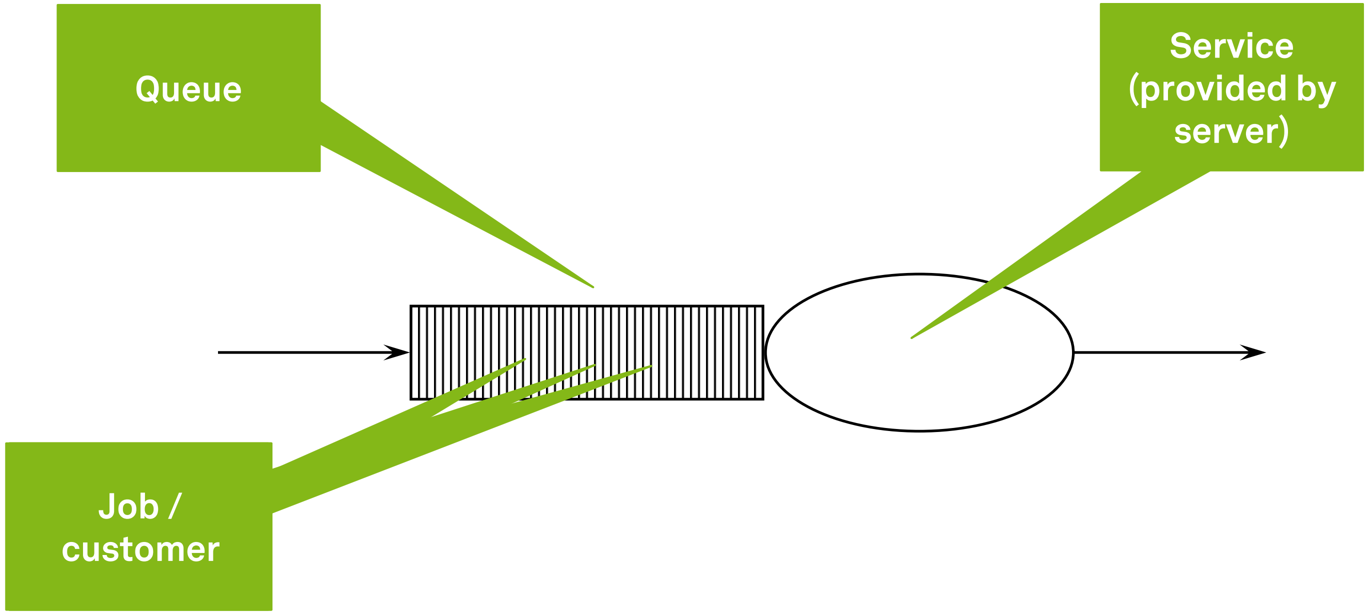
- **Capacity problems** are very common in industry and one of the main drivers of process redesign
  - Need to balance the cost of increased capacity against the gains of increased productivity and service
- Queuing and waiting time analysis is particularly important in **service systems**
  - Large costs of waiting and of lost sales due to waiting



Dumas et al. (2018)

Queues

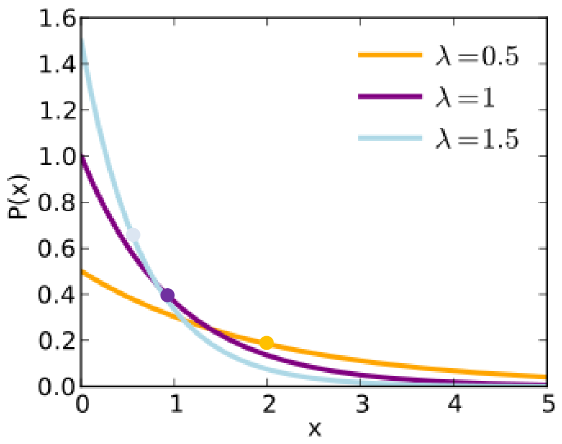
- Flow analysis **does not consider waiting times** due to resource contention
- **Queueing analysis** and **simulation** address these limitations and have a broader applicability

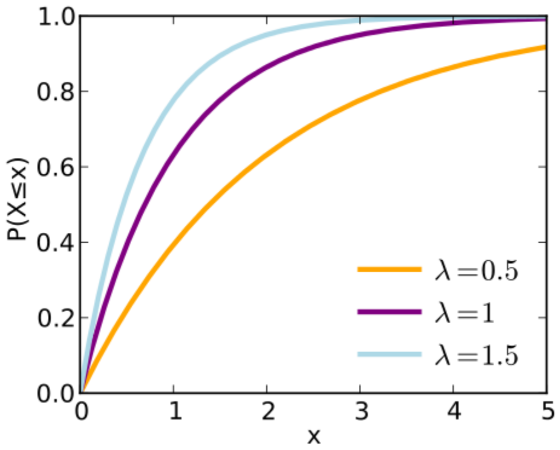


Dumas et al. (2018)

M/M/1 and M/M/c Queues

- Queue length in a system having 1 or *c* servers
- Arrivals and jobs are determined by a **Poisson process** (→ M)
  - **independent, identically distributed**, and (negative) **exponential**, uses **FIFO**





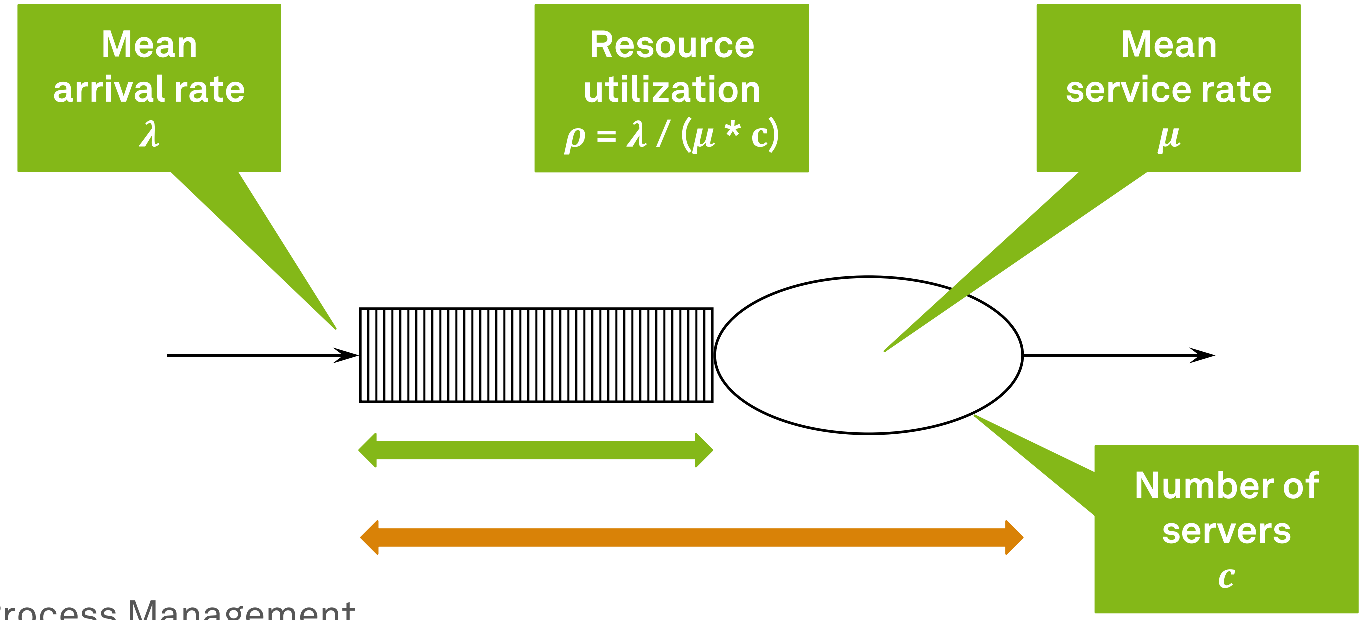
$$\lambda = (mean)\ arrival\ rate$$

$$\frac{1}{\lambda} = (mean)\ inter{-}arrival\ rate$$

Dumas et al. (2018)

M/M/1 and M/M/c Systems

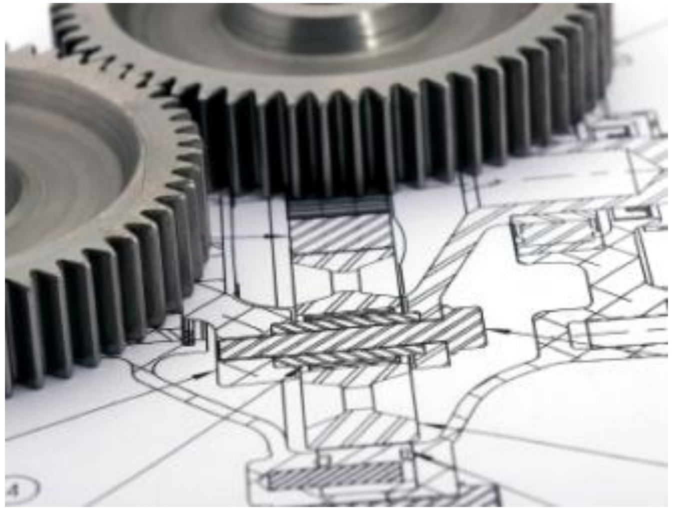
$$W_q = average\ time\ in\ queue\ (i.e. waiting\ time)$$
$$L_q = average\ number\ in\ queue\ (i.e. length\ of\ queue)$$
$$W = average\ time\ in\ system\ (i.e. cycle\ time)$$
$$L = average\ number\ in\ system\ (i.e. work\ in\ progress)$$



Dumas et al. (2018)

Example: Build-to-order

- *“Every 20 days, I receive 1 order.”*
- *“Every 10 days, I complete 1 order.”*
- *“What is my waiting time and my cycle time?”*
- *“What is my length of queue and my work-in-progress?”*



M/M/1 Systems

$$\lambda = 0{,}05\ (per\ day)$$
$$\frac{1}{\lambda} = 20\ (every\ 20\ days)$$
$$\mu = 0{,}1\ (per\ day)$$
$$\frac{1}{\mu} = 10\ (every\ 10\ days)$$

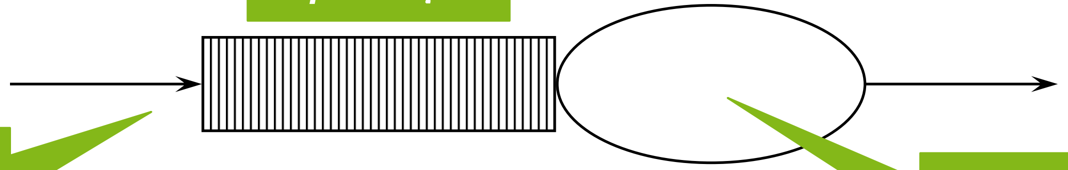
$$\rho = 0{,}5\ (50\ \%)$$

Length of queue

$$W_q = \frac{L_q}{\lambda}$$
$$L_q = \frac{\rho^2}{(1-\rho)}$$
$$W = W_q + \frac{1}{\mu}$$
$$L = \lambda W$$

$0{,}5^2 / (1 - 0{,}5) = 0{,}5$ jobs

Occupation rate
$\rho = \lambda / \mu$



Mean arrival rate
λ

Mean service rate
μ

Dumas et al. (2018)

M/M/1 Systems

$\lambda = 0{,}05\ (per\ day)$

$\frac{1}{\lambda} = 20\ (every\ 20\ days)$

$\mu = 0{,}1\ (per\ day)$

$\frac{1}{\mu} = 10\ (every\ 10\ days)$

Waiting time

$$W_q = \frac{L_q}{\lambda}$$

$$L_q = \frac{\rho^2}{(1-\rho)}$$

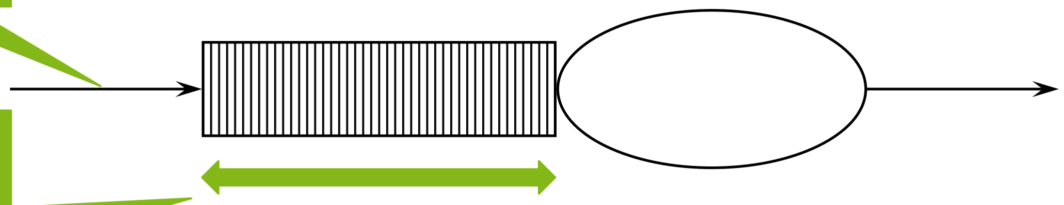
$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

0,5 / 0,05 = 10 days

$\rho = 0{,}5\ (50\ \%)$

$L_q = 0{,}5$



Dumas et al. (2018)

M/M/1 Systems

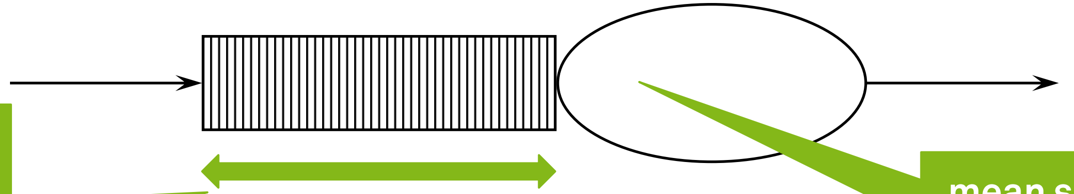
$$\lambda = 0{,}05\ (per\ day)$$
$$\frac{1}{\lambda} = 20\ (every\ 20\ days)$$
$$\mu = 0{,}1\ (per\ day)$$
$$\frac{1}{\mu} = 10\ (every\ 10\ days)$$

$$\rho = 0{,}5\ (50\ \%)$$
$$L_q = 0{,}5$$
$$W_q = 10$$

cycle time

$$W_q = \frac{L_q}{\lambda}$$
$$L_q = \frac{\rho^2}{(1-\rho)}$$
$$W = W_q + \frac{1}{\mu}$$
$$L = \lambda W$$

10 + 10 = 20 days



time in queue W_q

mean service rate μ

Dumas et al. (2018)

M/M/1 Systems

$\lambda = 0{,}05 \ (per\ day)$

$\frac{1}{\lambda} = 20 \ (every\ 20\ days)$

$\mu = 0{,}1 \ (per\ day)$

$\frac{1}{\mu} = 10 \ (every\ 10\ days)$

$\rho = 0{,}5 \ (50\ \%)$

$L_q = 0{,}5$

$W_q = 10$

$W = 20$

$$W_q = \frac{L_q}{\lambda}$$

$$L_q = \frac{\rho^2}{(1-\rho)}$$

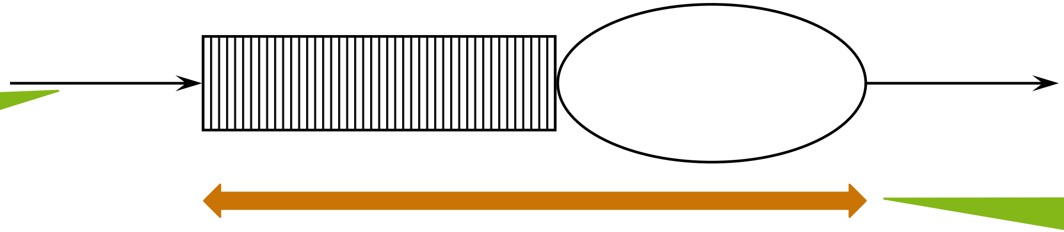
$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

Work-in-progress

0,05 * 20 = 1 job

Mean arrival rate λ



Time in system W

Dumas et al. (2018)

M/M/c Systems

$$W_q = \frac{L_q}{\lambda}$$

L_q is rather complex for M/M/c systems

$$W = W_q + \frac{1}{\mu}$$

$$L = \lambda W$$

$$L_q = \frac{(\lambda/\mu)^c \rho}{c!(1-\rho)^2 \left(\frac{(\lambda/\mu)^c}{c!(1-\rho)} + \sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} \right)}$$

Thank God, nobody does M/M/c without computers...

Dumas et al. (2018)

Summary Queueing Analysis

- Does not have to use negative exponential distribution

- **Limitations**
  - Generally not applicable when system includes parallel activities
    - One activity at a time
  - Requires case-by-case mathematical analysis
    - Queueing networks are available though
  - Assumes steady-state (valid only for long-term analysis)
  - Does not work for cost or quality measures

Dumas et al. (2018)

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Process Simulation

Process Simulation

- Uses **hypothetical instances** to create a **synthetic log**

- **Model** the process
- Enhance the process model with **simulation information**
  - Processing time (probability distribution for each task)
    - fixed / exponential (mean) / normal (mean + std dev)
  - Cost / value-add of tasks
  - Branching probabilities
  - Resource assignment

- **Run** the simulation
  - Define mean arrival rate (and distribution)
  - Define start date
  - Define end date / duration / number of instances

Dumas et al. (2018)

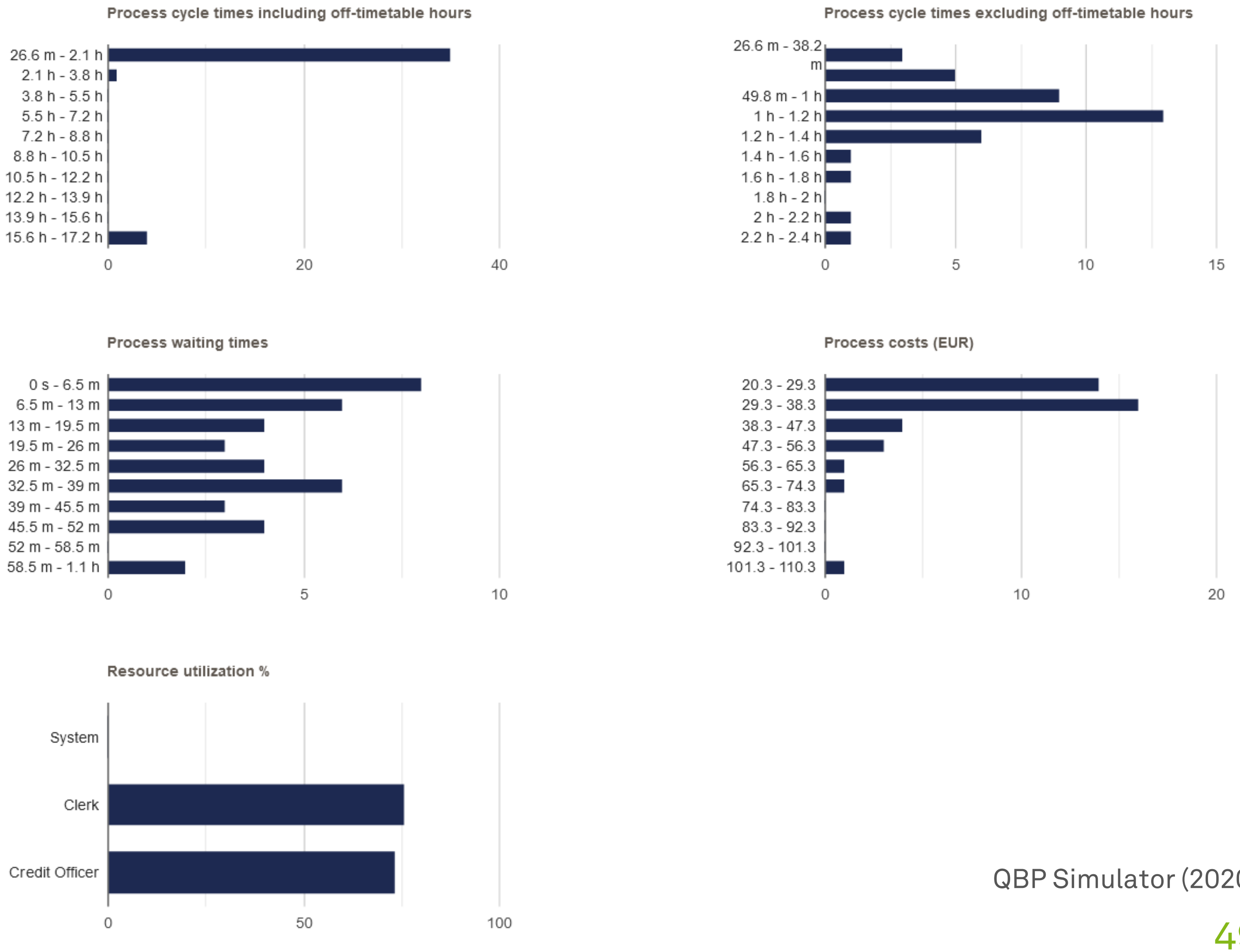
Process Simulation

- **Analyze** the simulation outputs
  - Process duration and cost stats and histograms
  - Waiting times (per activity)
  - Resource utilization (per resource)
- **Repeat** iteratively for alternative scenarios (caveat: stochasticity)
- **Cross-check** against expert advice
- **Limitations**
  - Results are based on models and **simplified** assumptions
  - Results depend on **accuracy** of input numbers, do sensitivity checks
  - **Human** resources are not robots, do credibility checks with stakeholders

Dumas et al. (2018)

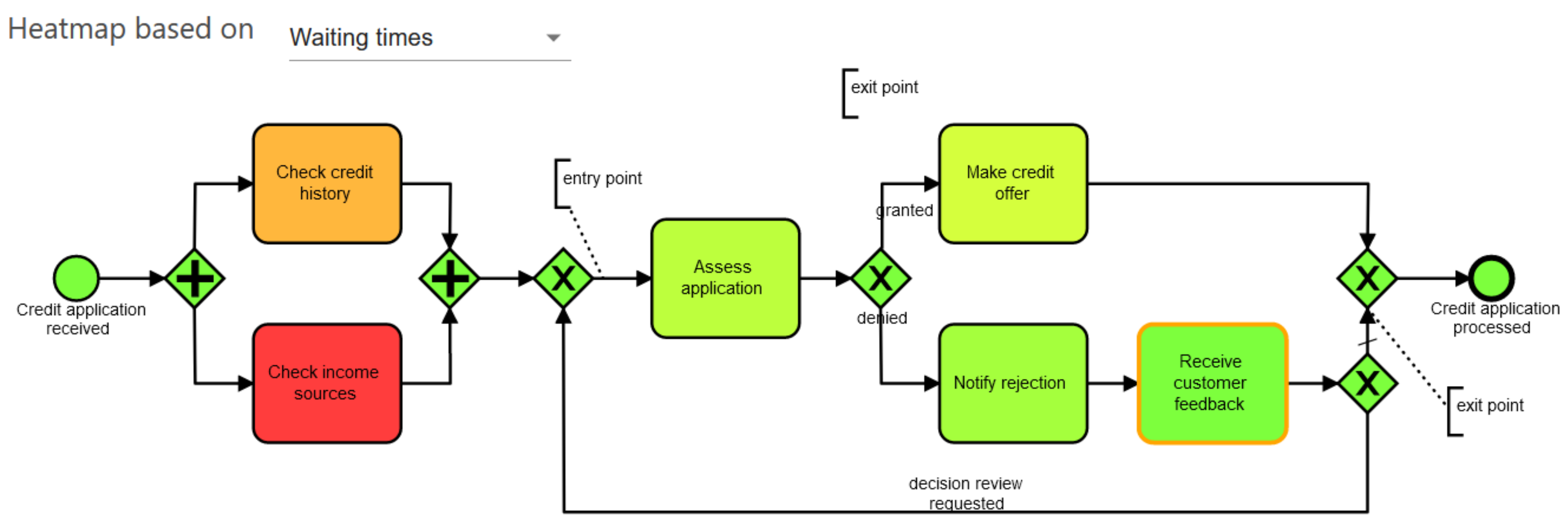
BIMP Example Output

- Other tools include
  - Appian
  - ARIS
  - IBM BPM
  - Oracle Business Process Analysis
  - Signavio Process Manager



QBP Simulator (2020)

BIMP Process Model Heatmap



Legend

Color	Value
	0 s
	1.3 m
	2.6 m
	3.9 m
	5.2 m
	6.6 m
	7.9 m
	9.2 m
	10.5 m
	11.8 m

QBP Simulator (2020)

Learning Goals of Today

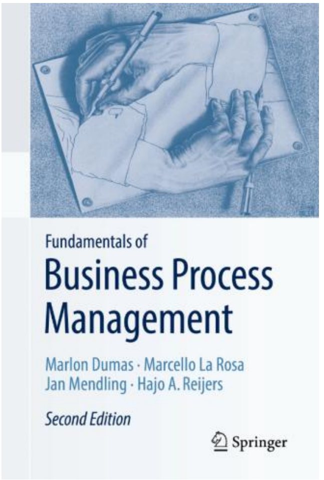
- Understand process performance measures
- Understand and be able to use flow analysis
- Understand and be able to use queueing analysis
- Understand process simulation

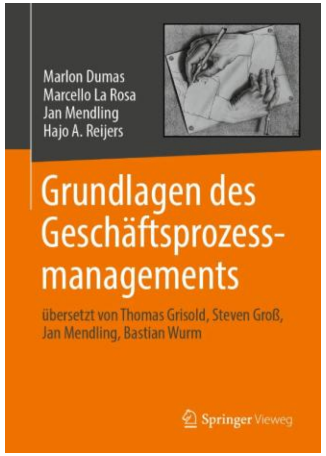
Recap Questions

- What are the four process performance dimensions of the Devil's Quadrangle? What are exemplary measures for each dimension and how could the quadrangle be extended in the future?
- What do we measure with flow analysis? What does it abstract from?
- What is resource contention and how can we measure it?

Literature

- **Section 2:** Process Identification
- **Section 7:** Quantitative Process Analysis

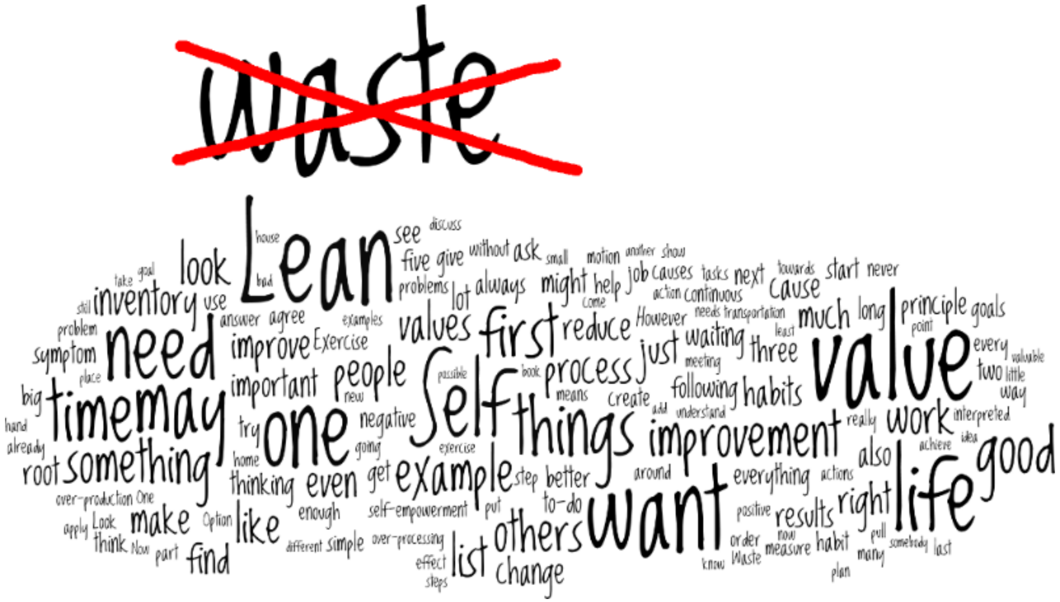




Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Any Questions about Quantitative Process Analysis?



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